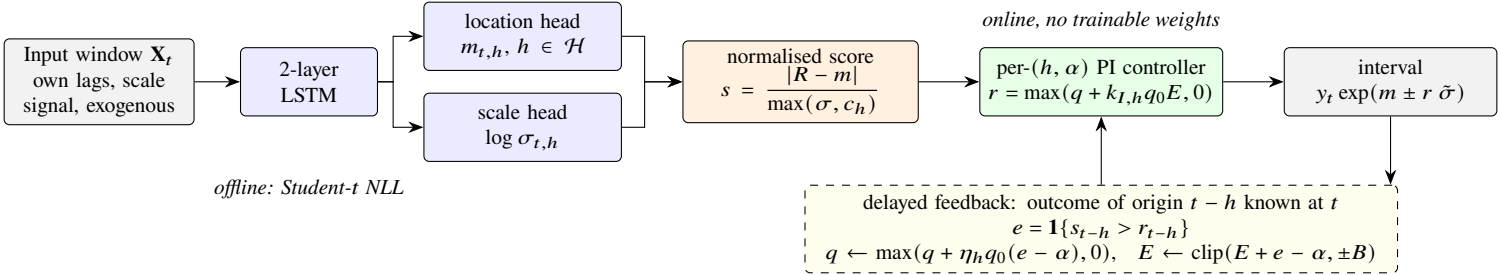
Input window $\mathbf{X}_t$
own lags, scale
signal, exogenous
2-layer
LSTM
location head
$m_{t,h},\ h \in \mathcal{H}$
scale head
$\log \sigma_{t,h}$
offline: Student-t NLL
normalised score
$s = \frac{|R - m|}{\max(\sigma, c_h)}$
online, no trainable weights
per-(h, α) PI controller
$r = \max(q + k_{I,h} q_0 E, 0)$
interval
$y_t \exp(m \pm r\, \tilde{\sigma})$
delayed feedback: outcome of origin $t - h$ known at t
$e = \mathbf{1}\{s_{t-h} > r_{t-h}\}$
$q \leftarrow \max(q + \eta_h q_0 (e - \alpha), 0), \quad E \leftarrow \mathrm{clip}(E + e - \alpha, \pm B)$